(bars: lengths by R, gain by a_{nom}; hats on estimates; $e_\theta = \theta - \hat{\theta}$; posterior quantities unless marked $^-$; $\bar{w}_d[k]$ commanded height, $\Delta\bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$; $\bar{a}' := \bar{a}'_{\mathrm{true}}(\bar{w}_d[k] - \delta\hat{\bar{w}}_3[k])$, $\bar{a}'' := \bar{a}''_{\mathrm{true}}(\bar{w}_d[k] - \delta\hat{\bar{w}}_3[k])$, both exogenous; $\Delta\bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$; $\Delta\hat{\bar{w}}[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\big[\delta\hat{\bar{w}}_3[k] + y_1[k] - \delta\hat{\bar{w}}_1[k]\big]$; $e_{\Delta\bar{w}} := \Delta\bar{w} - \Delta\hat{\bar{w}}$; $\mathbb{E}[\cdot]$ mean over noise realizations conditional on the data up to k; $P_{ij} = \mathbb{E}[e_i e_j]$; $d = 2$)

1 True dynamics

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\, \delta\bar{w}_3[k] - F_{dw}[k]\, e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\, \delta\bar{w}_3[k] + F_{dw}[k]\, e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] - \bar{a}_w[k] &= \bar{a}'_{\mathrm{true}}\big(\bar{w}[k]\big)\, \Delta\bar{w}[k] + \tfrac{1}{2}\, \bar{a}''_{\mathrm{true}}\big(\bar{w}[k]\big)\, \Delta\bar{w}[k]^2 + \mathcal{O}\big(\Delta\bar{w}^3\big) \\
\bar{a}'_{\mathrm{true}}\big(\bar{w}[k]\big) &= \bar{a}'_{\mathrm{true}}\big(\bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] - e_{\delta\bar{w}_3}[k]\big) = \bar{a}' - \bar{a}''\, e_{\delta\bar{w}_3}[k] + \mathcal{O}\big(e^2\big) \\
\bar{a}''_{\mathrm{true}}\big(\bar{w}[k]\big) &= \bar{a}'' + \mathcal{O}\big(e\big)
\end{aligned}
$$

$$
\bar{a}_w[k+1] - \bar{a}_w[k] = \big(\bar{a}' - \bar{a}''\, e_{\delta\bar{w}_3}[k]\big)\, \Delta\bar{w}[k] + \tfrac{1}{2}\, \bar{a}''\, \Delta\bar{w}[k]^2
$$

$$
\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1-\lambda_c) \sum_{i=1}^{d} \bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k]\, \bar{f}_T[k] + (1-\lambda_c) \sum_{i=1}^{d} \bar{a}_w[k-i]\, \bar{f}_T[k-i] + (1-\lambda_c)\, n_w[k] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k]
\end{aligned}
$$

$$
\begin{aligned}
&\mathbb{E}\big[\varepsilon_w[k]\, n_w[k]\big] = (1-\lambda_c)\, R_1 \\
&\qquad \tilde{\varepsilon}_w[k] := \varepsilon_w[k] - (1-\lambda_c)\, n_w[k], \qquad \mathbb{E}\big[\tilde{\varepsilon}_w\, n_w\big] = 0, \qquad \mathbb{E}\big[\tilde{\varepsilon}_w^2\big] = Q_{33} - (1-\lambda_c)^2 R_1 =: \tilde{Q}_{33} \\
&\qquad \Delta\bar{w}[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\big[\delta\bar{w}_3[k] + n_w[k]\big] + F_{dw}[k]\, e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]
\end{aligned}
$$

2 Estimator

$$
\begin{aligned}
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\, \delta\hat{\bar{w}}_3[k] - (1-\lambda_c)\big[y_1[k] - \delta\hat{\bar{w}}_1[k]\big] \\
\Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\big[\delta\hat{\bar{w}}_3[k] + y_1[k] - \delta\hat{\bar{w}}_1[k]\big] \\
\hat{\bar{a}}_w[k+1] - \hat{\bar{a}}_w[k] &= \bar{a}'\, \Delta\hat{\bar{w}}[k] + \tfrac{1}{2}\, \bar{a}''\, \Delta\hat{\bar{w}}[k]^2
\end{aligned}
$$

3 Error increment

$$
\begin{aligned}
y_1[k] - \delta\hat{\bar{w}}_1[k] &= e_{\delta\bar{w}_1}[k] + n_w[k] \\
e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k] = (1-\lambda_c)\big[e_{\delta\bar{w}_3}[k] - e_{\delta\bar{w}_1}[k]\big] + F_{dw}[k]\, e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]
\end{aligned}
$$

$$
\begin{aligned}
e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] &= \big(\bar{a}_w[k+1] - \bar{a}_w[k]\big) - \big(\hat{\bar{a}}_w[k+1] - \hat{\bar{a}}_w[k]\big) \\
&= \big(\bar{a}' - \bar{a}''\, e_{\delta\bar{w}_3}\big)\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big) + \tfrac{1}{2}\,\bar{a}''\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big)^2 - \bar{a}'\,\Delta\hat{\bar{w}} - \tfrac{1}{2}\,\bar{a}''\,\Delta\hat{\bar{w}}^2 \\
&= \bar{a}'\, e_{\Delta\bar{w}} - \bar{a}''\, e_{\delta\bar{w}_3}\,\Delta\hat{\bar{w}} - \bar{a}''\, e_{\delta\bar{w}_3}\, e_{\Delta\bar{w}} + \bar{a}''\,\Delta\hat{\bar{w}}\, e_{\Delta\bar{w}} + \tfrac{1}{2}\,\bar{a}''\, e_{\Delta\bar{w}}^2 + \mathcal{O}(3)
\end{aligned}
$$

4 Mean

$$
\begin{aligned}
\mathbb{E}[e_{\delta\bar{w}_1}] = \mathbb{E}[e_{\delta\bar{w}_3}] = \mathbb{E}[e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w] &= 0 \\
\mathbb{E}[e_{\delta\bar{w}_3}^2] = P_{33}, \quad \mathbb{E}[e_{\delta\bar{w}_1}^2] = P_{11}, \quad \mathbb{E}[e_{\delta\bar{w}_1} e_{\delta\bar{w}_3}] &= P_{13} \\
\mathbb{E}[e_{\delta\bar{w}_3}\, e_{\bar{a}_w}] = P_{34}, \quad \mathbb{E}[e_{\delta\bar{w}_1}\, e_{\bar{a}_w}] = P_{14}, \quad \mathbb{E}[e_{\bar{a}_w}^2] &= P_{44} \\
\mathbb{E}[\tilde{\varepsilon}_w\, e_{\delta\bar{w}_1}] = \mathbb{E}[\tilde{\varepsilon}_w\, e_{\delta\bar{w}_3}] = \mathbb{E}[\tilde{\varepsilon}_w\, e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w\, n_w] &= 0 \\
\mathbb{E}\big[e_j\,\delta\hat{\bar{w}}_i\big] = \mathbb{E}\big[e_j\, y_1\big] &= 0 \qquad \text{(posterior error $\perp$ data)}
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}\big[\bar{a}'\, e_{\Delta\bar{w}}\big] &= 0 \\
\mathbb{E}\big[-\bar{a}''\, e_{\delta\bar{w}_3}\,\Delta\hat{\bar{w}}\big] &= 0 \\
\mathbb{E}\big[-\bar{a}''\, e_{\delta\bar{w}_3}\, e_{\Delta\bar{w}}\big] &= -\bar{a}''\,\big[(1-\lambda_c)\big(P_{33} - P_{13}\big) + F_{dw} P_{34}\big] \\
\mathbb{E}\big[\bar{a}''\,\Delta\hat{\bar{w}}\, e_{\Delta\bar{w}}\big] &= 0 \\
\mathbb{E}\big[\tfrac{1}{2}\,\bar{a}''\, e_{\Delta\bar{w}}^2\big] &= \tfrac{1}{2}\,\bar{a}''\,\big[(1-\lambda_c)^2\big(P_{33} + P_{11} - 2P_{13}\big) + 2(1-\lambda_c)F_{dw}\big(P_{34} - P_{14}\big) \\
&\qquad + F_{dw}^2 P_{44} + \tilde{Q}_{33}\big]
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] = \bar{a}''\Big\{ &-(1-\lambda_c)\big(P_{33} - P_{13}\big) - F_{dw} P_{34} \\
&+ \tfrac{1}{2}(1-\lambda_c)^2\big(P_{33} + P_{11} - 2P_{13}\big) + (1-\lambda_c)F_{dw}\big(P_{34} - P_{14}\big) \\
&+ \tfrac{1}{2}F_{dw}^2 P_{44} + \tfrac{1}{2}\tilde{Q}_{33}\Big\}
\end{aligned}
$$

(first, second and fourth lines: $\bar{a}'$ and $\Delta\hat{\bar{w}}$ are functions of the data, $e_{\Delta\bar{w}}$ and $e_{\delta\bar{w}_3}$ are posterior errors, $\mathbb{E}[\,e\,|\,\text{data}\,] = 0$; without the $y_1 - \delta\hat{\bar{w}}_1$ input $e_{\Delta\bar{w}}$ contains $(1-\lambda_c)n_w$, which is correlated with the data, and these means become the n_w feedthrough shares of 0902_formC_aptrue_4state.tex, Error dynamics, each $\propto \bar{a}''(1-\lambda_c)\ell_{31}R_1$, whose sum was measured to vanish)

5 Estimator with the mean term

$$
\begin{aligned}
\hat{\bar{a}}_w[k+1] = \hat{\bar{a}}_w[k] + \bar{a}'\,\Delta\hat{\bar{w}}[k] + \tfrac{1}{2}\,\bar{a}''\,\Delta\hat{\bar{w}}[k]^2 \\
+ \bar{a}''\Big\{ -(1-\lambda_c)\big(P_{33}[k] - P_{13}[k]\big) - F_{dw}[k]\,P_{34}[k] \\
+ \tfrac{1}{2}(1-\lambda_c)^2\big(P_{33}[k] + P_{11}[k] - 2P_{13}[k]\big) \\
+ (1-\lambda_c)F_{dw}[k]\big(P_{34}[k] - P_{14}[k]\big) + \tfrac{1}{2}F_{dw}[k]^2 P_{44}[k] + \tfrac{1}{2}\tilde{Q}_{33}[k]\Big\}
\end{aligned}
$$

$$\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] = 0$$

(the braced term is a deterministic input to the predict; it is not differentiated and does not enter F_e, H, Q)

6 Jacobians

$$
\begin{aligned}
&\frac{\partial\,\delta\hat{\bar{w}}_3[k+1]}{\partial\,\delta\hat{\bar{w}}_3[k]} = \lambda_c, \qquad \frac{\partial\,\delta\hat{\bar{w}}_3[k+1]}{\partial\,\delta\hat{\bar{w}}_1[k]} = (1-\lambda_c) \\
&\frac{\partial\hat{\bar{a}}_w[k+1]}{\partial\delta\hat{\bar{w}}_3[k]} = (1-\lambda_c)\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) - \bar{a}''\,\Delta\hat{\bar{w}}[k] \;\to\; (1-\lambda_c)\,\bar{a}' \\
&\frac{\partial\hat{\bar{a}}_w[k+1]}{\partial\delta\hat{\bar{w}}_1[k]} = -(1-\lambda_c)\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) \;\to\; -(1-\lambda_c)\,\bar{a}' \\
&\frac{\partial\bar{a}_w[k+1]}{\partial\bar{a}_w[k]} = 1 + F_{dw}[k]\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) \;\to\; 1 + F_{dw}[k]\,\bar{a}'
\end{aligned}
$$

(limits at $\Delta\hat{\bar{w}} \to 0$, the tangent the code keeps; the $-\bar{a}''\Delta\hat{\bar{w}}$ in the first gain-row entry is $\partial\bar{a}'/\partial\delta\hat{\bar{w}}_3 = -\bar{a}''$ of the 'est' reading; the last entry and $\partial\,\delta\bar{w}_3[k+1]/\partial\bar{a}_w[k] = -F_{dw}$ are entries of the TRUE transition, not of the estimator's lines (which do not contain $\hat{\bar{a}}_w$): the truth's $F_{dw}\,e_{\bar{a}_w}$ inside $\Delta\bar{w}$, written as $F_{dw}(\bar{a}_w - \hat{\bar{a}}_w)$)

7 Error dynamics

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] + (1-\lambda_c)\,e_{\delta\bar{w}_1}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \tilde{\varepsilon}_w[k] \\
e_{\bar{a}_w}[k+1] &= -(1-\lambda_c)\,\bar{a}'\,e_{\delta\bar{w}_1}[k] + (1-\lambda_c)\,\bar{a}'\,e_{\delta\bar{w}_3}[k] + \big(1 + F_{dw}\,\bar{a}'\big)\,e_{\bar{a}_w}[k] + \bar{a}'\,\tilde{\varepsilon}_w[k] \\
&\quad - \bar{a}''\,e_{\delta\bar{w}_3}\,\Delta\hat{\bar{w}} - \bar{a}''\,e_{\delta\bar{w}_3}\,e_{\Delta\bar{w}} + \bar{a}''\,\Delta\hat{\bar{w}}\,e_{\Delta\bar{w}} + \tfrac{1}{2}\,\bar{a}''\,e_{\Delta\bar{w}}^2 - \mathbb{E}[\,\cdot\,]
\end{aligned}
$$

(third line: $\delta\bar{w}_3[k+1] - \delta\hat{\bar{w}}_3[k+1]$ with $y_1 - \delta\hat{\bar{w}}_1 = e_{\delta\bar{w}_1} + n_w$, the n_w of ε_w cancels; fourth line: the second-order group minus its mean of the previous section)

$$\mathbf{e}[k+1] = \tilde{F}_e[k]\,\mathbf{e}[k] + \tilde{\mathbf{q}}[k], \qquad \tilde{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (1-\lambda_c) & 0 & \lambda_c & -F_{dw} \\ -(1-\lambda_c)\,\bar{a}' & 0 & (1-\lambda_c)\,\bar{a}' & 1 + F_{dw}\,\bar{a}' \end{bmatrix}$$

$$\tilde{F}_e[k] = F_e[k] - g_n[k]\,\mathbf{e}_1^\top, \qquad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} \\ 0 & 0 & (1-\lambda_c)\,\bar{a}' & 1 + F_{dw}\,\bar{a}' \end{bmatrix}$$

$$g_n[k] = (1-\lambda_c)\begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \qquad \tilde{\mathbf{q}}[k] = \tilde{\varepsilon}_w[k]\begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \qquad \mathbb{E}\big[\tilde{\mathbf{q}}[k]\,n_w[k]\big] = 0$$

$$P^f[k+1] = \tilde{F}_e[k]\,P[k]\,\tilde{F}_e[k]^\top + \tilde{Q}[k]$$

8 Measurement and update

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k], \qquad \bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}'\,\nabla_d \bar{w}_d[k] \\
H &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
e_{y_1}[k] &= e^-_{\delta\bar{w}_1}[k] + n_w[k], \qquad e_{y_2}[k] = e^-_{\bar{a}_w}[k] + n_a[k]
\end{aligned}$$

$$\begin{aligned}
L[k] &= P^f H^\top \big(H P^f H^\top + R\big)^{-1} \\
\hat{\mathbf{x}}[k] &= \hat{\mathbf{x}}^-[k] + L[k]\,\mathbf{e}_y[k] \\
P[k] &= \big(I - L[k]H\big)P^f[k] \\
y_1[k] - \delta\hat{w}_1[k] &= \frac{R_1}{P^f_{11}[k] + R_1}\,e_{y_1}[k] - \ell_{12}[k]\,\big(e_{y_2}[k] - \ell_{41}[k]\,e_{y_1}[k]\big) = \mathbb{E}\big[n_w[k]\,\big|\,y[1..k]\big]
\end{aligned}$$

(last line: the posterior residual the Estimator section feeds into the next step, written for the sequential update y_1 then y_2 as in the code; $e_{y_2} - \ell_{41}e_{y_1}$ is the y_2 innovation after the y_1 update, $\ell_{12} = H_{24}P^{(1)}_{14}/S_2$ equals the joint L_{12}; exact in the linear-Gaussian model, approximate for the EKF)

9 Noise

$$
\begin{aligned}
\kappa_T &= \frac{4k_B T\, a_o}{R}, \qquad \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2} \\
\tilde{Q}_{33} &= \kappa_T \Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2 \big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2] \big) \Big) \qquad \big(= Q_{33} - (1-\lambda_c)^2 \sigma_{n_w}^2 \big) \\
\tilde{Q}_{34} = \tilde{Q}_{43} &= -\bar{a}'\, \tilde{Q}_{33} \\
\tilde{Q}_{44} &= \bar{a}'^2\, \tilde{Q}_{33} \\
R &= \mathrm{diag}(R_1, R_2), \qquad R_1 = \sigma_{n_w}^2, \qquad R_2 = 2a_{\mathrm{cov}}^2\, IF_{\mathrm{var}}\big(\hat{\bar{a}}_w\big)\big(\hat{\bar{a}}_w + \xi\big)^2
\end{aligned}
$$

(IF_{var}, ξ as in 0902_formC_aptrue_4state.tex; the n_w share $(1-\lambda_c)^2\sigma_{n_w}^2$ has left Q and entered the Estimator section as the input $g_n\,(y_1 - \delta\hat{\bar{w}}_1)$)

10 b_{true} arm: the law read at the estimated gain

(this arm: $b := b_{\text{true}}(\bar{w}[k-1])$ read at the previous step's TRUE height (code, b_true_at = 'true'), treated as exogenous; slot 5 locked; $\bar{a}' := b\,\big(1-\hat{\bar{a}}_w[k]\big)^2$ and $\bar{a}'' := -2b\big(1-\hat{\bar{a}}_w[k]\big)\,\bar{a}'$ read at the ESTIMATED GAIN, not at a height; $\dfrac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w} = -2b\big(1-\hat{\bar{a}}_w\big) = \dfrac{\bar{a}''}{\bar{a}'}$; everything not listed is as in Sections 1–9)

True dynamics, the slope line

$$\bar{a}'_{\text{true}}\big(\bar{w}[k]\big) = b\,\big(1-\bar{a}_w[k]\big)^2 = b\,\big(1-\hat{\bar{a}}_w[k]-e_{\bar{a}_w}[k]\big)^2 = \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,e_{\bar{a}_w}[k] + \mathcal{O}\big(e^2\big)$$

$$\bar{a}_w[k+1]-\bar{a}_w[k] = \Big(\bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,e_{\bar{a}_w}[k]\Big)\,\Delta\bar{w}[k] + \tfrac{1}{2}\,\bar{a}''\,\Delta\bar{w}[k]^2$$

Error increment (the $-\bar{a}''\,e_{\delta\bar{w}_3}$ lines of Section 3 are replaced)

$$e_{\bar{a}_w}[k+1]-e_{\bar{a}_w}[k] = \bar{a}'\,e_{\Delta\bar{w}} + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,e_{\bar{a}_w}\,\Delta\hat{\bar{w}} + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,e_{\bar{a}_w}\,e_{\Delta\bar{w}} + \bar{a}''\,\Delta\hat{\bar{w}}\,e_{\Delta\bar{w}} + \tfrac{1}{2}\,\bar{a}''\,e^2_{\Delta\bar{w}} + \mathcal{O}(3)$$

Mean

$$\begin{aligned}
\mathbb{E}\Big[\frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,e_{\bar{a}_w}\,\Delta\hat{\bar{w}}\Big] &= 0\\
\mathbb{E}\Big[\frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,e_{\bar{a}_w}\,e_{\Delta\bar{w}}\Big] &= \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\big[(1-\lambda_c)\big(P_{34}-P_{14}\big)+F_{dw}P_{44}\big]\\
\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big]-\mathbb{E}\big[e_{\bar{a}_w}[k]\big] &= \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\big[(1-\lambda_c)\big(P_{34}-P_{14}\big)+F_{dw}P_{44}\big]\\
&\quad+\tfrac{1}{2}\,\bar{a}''\,\big[(1-\lambda_c)^2\big(P_{33}+P_{11}-2P_{13}\big)+2(1-\lambda_c)F_{dw}\big(P_{34}-P_{14}\big)\\
&\qquad\quad+F_{dw}^2P_{44}+\tilde{Q}_{33}\big]
\end{aligned}$$

Estimator with the mean term

$$\begin{aligned}
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \bar{a}'\,\Delta\hat{\bar{w}}[k] + \tfrac{1}{2}\,\bar{a}''\,\Delta\hat{\bar{w}}[k]^2\\
&\quad+\frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\big[(1-\lambda_c)\big(P_{34}[k]-P_{14}[k]\big)+F_{dw}[k]\,P_{44}[k]\big]\\
&\quad+\tfrac{1}{2}\,\bar{a}''\,\big[(1-\lambda_c)^2\big(P_{33}[k]+P_{11}[k]-2P_{13}[k]\big)+2(1-\lambda_c)F_{dw}[k]\big(P_{34}[k]-P_{14}[k]\big)\\
&\qquad\quad+F_{dw}[k]^2P_{44}[k]+\tilde{Q}_{33}[k]\big]
\end{aligned}$$

$$\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big]-\mathbb{E}\big[e_{\bar{a}_w}[k]\big] = 0$$

Difference to the Section 5 term (gain-read minus height-read start point)

$$\begin{aligned}
&\frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\big[(1-\lambda_c)\big(P_{34}-P_{14}\big)+F_{dw}P_{44}\big]+\bar{a}''\,\big[(1-\lambda_c)\big(P_{33}-P_{13}\big)+F_{dw}P_{34}\big]\\
&\quad=\frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\Big[(1-\lambda_c)\big(P_{34}+\bar{a}'P_{33}-P_{14}-\bar{a}'P_{13}\big)+F_{dw}\big(P_{44}+\bar{a}'P_{34}\big)\Big]\\
&\quad=\frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\mathbb{E}\Big[\big(e_{\bar{a}_w}+\bar{a}'\,e_{\delta\bar{w}_3}\big)\,e_{\Delta\bar{w}}\Big]
\end{aligned}$$

(= 0 when the gain error is slaved to the position error through the law, $e_{\bar{a}_w} = -\bar{a}'\,e_{\delta\bar{w}_3}$, i.e. $P_{34} = -\bar{a}'P_{33}$, $P_{14} = -\bar{a}'P_{13}$ and $P_{44} = \bar{a}'^2P_{33}$ (the first two alone leave $F_{dw}(P_{44}-\bar{a}'^2P_{33})$); hold, both trajectories: $P_{34} + \bar{a}' P_{33} \approx 3 \times 10^{-9}$, the difference is $< 0.005 \times 10^{-6}$ per step; near-wall half of the descent: -3.5×10^{-6} (canon) and -0.5×10^{-6} (Meng) per step in $\mathbb{E}[\hat{\bar{a}}_w - \bar{a}_w]$, running sums -0.0026 and -0.0040 at the start of the hold (code: ctrl_const.pred_mean2_e4, the difference of the two pred_mean2 logs, 10 seeds, run_btrue_e4.m); the closed-loop response to this input is smaller than its running sum, $+0.0010$ (canon) and $+0.0004$ (Meng) at the end of the descent, relaxing to < 0.0003 within the following second)

(treating b as exogenous drops the mean $-(1-\hat{\bar{a}}_w)^2\, b'_{\text{true}}\, \mathbb{E}\big[e_{\delta\bar{w}_3}[k-1]\, e_{\Delta\bar{w}}\big]$ of $\bar{a}'\, e_{\Delta\bar{w}}$, b being read at the true height $\bar{w}_d[k-1] - \delta\bar{w}_3[k-1]$; relative to the kept $\bar{a}''$ term it is $\sim b'_{\text{true}}/\big(2b^2(1-\hat{\bar{a}}_w)\big)$, a few percent, b_{true} spanning 8/9 to 1 over the band)

Jacobians and error dynamics (the entries that change)

$$\frac{\partial \hat{\bar{a}}_w[k+1]}{\partial \delta\hat{\bar{w}}_3[k]} = (1-\lambda_c)\big(\bar{a}' + \bar{a}''\, \Delta\hat{\bar{w}}[k]\big) \;\to\; (1-\lambda_c)\, \bar{a}'$$
$$\frac{\partial \hat{\bar{a}}_w[k+1]}{\partial \hat{\bar{a}}_w[k]} = 1 + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, \Delta\hat{\bar{w}}[k] + F_{dw}[k]\big(\bar{a}' + \bar{a}''\, \Delta\hat{\bar{w}}[k]\big) \;\to\; 1 + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, \Delta\hat{\bar{w}}[k] + F_{dw}[k]\, \bar{a}'$$
$$\tilde{F}_e(4,4) = 1 + F_{dw}\, \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, \Delta\hat{\bar{w}}$$
$$y_2[k] = \bar{a}_w[k] - \bar{a}'\, \nabla_d \bar{w}_d[k] + n_a[k], \qquad H_{24} = 1 - \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, \nabla_d \bar{w}_d[k]$$

(no $-\bar{a}''\Delta\hat{\bar{w}}$ in the $\delta\hat{\bar{w}}_3$ entry: the slope does not read a height; $H_{24} \neq 1$ whenever $\bar{a}'$ reads $\hat{\bar{a}}_w$ (Sections 10, 11; code line `1 - Grad_wbar_d*A_a_i`), $= 1$ only for the exogenous slope of Sections 1–9; the $\partial\bar{a}'/\partial\hat{\bar{a}}_w\, \Delta\hat{\bar{w}}$ entry is the law's own amplification of a gain error along the step, > 1 when moving toward the wall; $\tilde{Q}_{34} = -\bar{a}'\tilde{Q}_{33}$, $\tilde{Q}_{44} = \bar{a}'^2\tilde{Q}_{33}$ unchanged)

11 Production: b a constant state

(this arm: $\hat{b} := \hat{x}_5$, constant model, $\tilde{Q}_{55} = 0$, seed 8/9, $\sqrt{P_{55}[0]} = \sup_{\text{env}} |b_{\text{true}} - 8/9|$; $\bar{a}' := \hat{b}\,(1-\hat{\bar{a}}_w)^2$, $\dfrac{\partial \bar{a}'}{\partial \hat{b}} = (1-\hat{\bar{a}}_w)^2 = \dfrac{\bar{a}'}{\hat{b}}$, $\bar{a}'' := -2\hat{b}(1-\hat{\bar{a}}_w)\,\bar{a}'$; $e_b[k] := b_{\text{true}}\big(\bar{w}[k]\big) - \hat{b}[k]$, NOT zero-mean: b_{true} is a curve, $\hat{b}$ a constant; $f(\bar{w}) := \big(1 - \bar{a}_{\text{true}}(\bar{w})\big)^2\big(b_{\text{true}}(\bar{w}) - \hat{b}\big)$, $F' = f$; everything not listed is as in Sections 1–10)

True dynamics, slope and curvature

$$\bar{a}'_{\text{true}}\big(\bar{w}[k]\big) = b_{\text{true}}\big(\bar{w}[k]\big)\big(1-\bar{a}_w[k]\big)^2 = \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, e_{\bar{a}_w}[k] + \frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b[k] + \mathcal{O}\big(e^2\big)$$
$$\bar{a}''_{\text{true}}\big(\bar{w}[k]\big) = \frac{d}{d\bar{w}}\Big[b_{\text{true}}(\bar{w})\big(1-\bar{a}_{\text{true}}(\bar{w})\big)^2\Big] = \bar{a}'' + f'\big(\bar{w}[k]\big) + \frac{\bar{a}''}{\bar{a}'}\, f\big(\bar{w}[k]\big) + \mathcal{O}(e_{\bar{a}_w})$$

($\bar{a}''/\bar{a}' = -2\hat{b}(1-\hat{\bar{a}}_w)$; the $(\bar{a}''/\bar{a}')f$ piece is first order in e_b, the same order as the model-error line, and is kept)

$\hat{b}$ and e_b

$$\begin{aligned}
\hat{b}^-[k+1] &= \hat{b}[k] \\
\hat{b}[k+1] &= \hat{b}^-[k+1] + \ell_{51}[k+1]\, e_{y_1}[k+1] + \ell_{52}[k+1]\, e_{y_2}[k+1] \\
e_b[k+1] &= e_b[k] + b'_{\text{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k] - \ell_{51}[k+1]\, e_{y_1}[k+1] - \ell_{52}[k+1]\, e_{y_2}[k+1]
\end{aligned}$$

Error increment (Section 10 plus the e_b and f' lines)

$$\begin{aligned}
e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] = \bar{a}'\, e_{\Delta\bar{w}} &+ \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, e_{\bar{a}_w}\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big) + \frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big) + \bar{a}''\,\Delta\hat{\bar{w}}\, e_{\Delta\bar{w}} + \tfrac{1}{2}\,\bar{a}''\, e_{\Delta\bar{w}}^2 \\
&+ \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'}\, f\Big)\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big)^2 + \mathcal{O}(3)
\end{aligned}$$

Mean, the e_b and curvature lines ($P_{i5} :=$ the filter's own posterior cross-covariance under its constant-b model; the part of e_b that fluctuates given the data is $-b'_{\text{true}}\, e_{\delta\bar{w}_3}$, which is NOT P_{i5})

$$\begin{aligned}
\mathbb{E}\Big[\frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b\,\Delta\hat{\bar{w}}\Big] &= \frac{\partial \bar{a}'}{\partial \hat{b}}\,\mathbb{E}[e_b]\,\Delta\hat{\bar{w}} \qquad \text{(first order; } \mathbb{E}[e_b] = b_{\text{true}}(\bar{w}) - \mathbb{E}[\hat{b}] \neq 0\text{)} \\
\mathbb{E}\Big[\frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b\, e_{\Delta\bar{w}}\Big] &= \frac{\partial \bar{a}'}{\partial \hat{b}}\,\big[(1-\lambda_c)\big(P_{35} - P_{15}\big) + F_{dw}P_{45}\big] \\
&\quad - \frac{\partial \bar{a}'}{\partial \hat{b}}\, b'_{\text{true}}\,\big[(1-\lambda_c)\big(P_{33} - P_{13}\big) + F_{dw}P_{34}\big] \\
\mathbb{E}\Big[\tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'} f\Big)\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big)^2\Big] &= \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'} f\Big)\big(\Delta\hat{\bar{w}}^2 + \mathbb{E}[e_{\Delta\bar{w}}^2]\big)
\end{aligned}$$

(second line: the first bracket is the filter's cross-covariance with its constant unknown b, the second is the height dependence of b_{true} through $e_{\delta\bar{w}_3}$; b'_{true} is not c-free)

$$\begin{aligned}
\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] = &\frac{\partial \bar{a}'}{\partial \hat{b}}\,\mathbb{E}[e_b]\,\Delta\hat{\bar{w}} + \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'} f\Big)\big(\Delta\hat{\bar{w}}^2 + \mathbb{E}[e_{\Delta\bar{w}}^2]\big) \\
&- \frac{\partial \bar{a}'}{\partial \hat{b}}\, b'_{\text{true}}\big[(1-\lambda_c)\big(P_{33} - P_{13}\big) + F_{dw}P_{34}\big] \\
&+ \frac{\partial \bar{a}'}{\partial \hat{b}}\,\big[(1-\lambda_c)\big(P_{35} - P_{15}\big) + F_{dw}P_{45}\big] + \text{(Section 10 lines)}
\end{aligned}$$

(first two lines: model error, needs b_{true} and b'_{true}, not c-free, not added; third line: added, the Section 10 term with e_b)

Estimator with the mean term (the line added to Section 10's)

$$\hat{\bar{a}}_w[k+1] = (\text{Section 10}) + \frac{\partial \bar{a}'}{\partial \hat{b}}\,\big[(1-\lambda_c)\big(P_{35}[k]-P_{15}[k]\big) + F_{dw}[k]\,P_{45}[k]\big]$$

Model error along a path (first order in e_b; open loop, $\mu := \mathbb{E}[e_{\bar{a}_w}]_{\text{model}}$)

$$\mu[k+1] = \Big(1 + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\Delta\bar{w}[k]\Big)\mu[k] + f\big(\bar{w}[k]\big)\,\Delta\bar{w}[k] + \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'}f\Big)\Delta\bar{w}[k]^2 + \mathcal{O}(3)$$

$$\frac{d\mu}{d\bar{w}} = \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\mu + f \qquad \Longrightarrow \qquad \mu(\bar{w}) = \big(1-\bar{a}_{\text{true}}(\bar{w})\big)^2 \int_{\bar{w}_{\text{start}}}^{\bar{w}} \big(b_{\text{true}} - \hat{b}\big)\,d\bar{w}'$$

$$\frac{1}{1-\bar{a}_{\text{true}}(\bar{w})} - \frac{1}{1-\hat{\bar{a}}_w(\bar{w})} = \int_{\bar{w}_{\text{start}}}^{\bar{w}} \big(b_{\text{true}} - \hat{b}\big)\,d\bar{w}' \qquad (\text{exact, noise-free: } \tfrac{d}{d\bar{w}}\tfrac{1}{1-\bar{a}} = b)$$

stationary hold: $\mathbb{E}\big[\Delta\mu\big] \to 0$

$$\mathbb{E}[f\,\Delta\bar{w}] = -(1-\lambda_c)\,f'\,\mathrm{Var}(\bar{w}), \qquad \tfrac{1}{2}f'\,\mathbb{E}[\Delta\bar{w}^2] = +(1-\lambda_c)\,f'\,\mathrm{Var}(\bar{w})$$

$$\tfrac{1}{2}\tfrac{\bar{a}''}{\bar{a}'}f\,\mathbb{E}[\Delta\bar{w}^2] + \tfrac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\mathbb{E}[\Delta\bar{w}\,e_{\bar{a}_w}] = 0 \qquad (\text{likewise})$$

(the forcing f alone integrates to $F(\bar{w}_{\text{end}}) - F(\bar{w}_{\text{start}})$, $F' = f$; the law then AMPLIFIES what was injected far away by $[(1-\bar{a}_{\text{wall}})/(1-\bar{a}_{\text{far}})]^2 = 29.8$ from $\bar{w} = 6.67$ to 1.10 – the open loop of the measured $\times 22$)

($\hat{b} = 8/9$, $c_\perp$ of the plant: $b_{\text{true}}(6.67) = 0.883$, minimum 0.867 at $\bar{w} = 2.19$, crosses 8/9 at $\bar{w} = 1.325$, $b_{\text{true}}(1.10) = 0.928$; $f(1.10) = +0.032$, $f(1.8) = -0.0066$, $f(6.67) = -0.0002$. Open loop, $\bar{a}_w - \hat{\bar{a}}_w$ at the wall: Meng starts at $\bar{w} = 6.67$ (15 μm): $\int_{6.67}^{1.10}(b_{\text{true}} - 8/9)\,d\bar{w} = +0.068$, $\mu(1.10) = (1-0.0873)^2 \times 0.068 = +0.057$ (exact law integration +0.060, 69% of $\bar{a}_w(1.10)$); canon starts at $\bar{w} = 22.2$ (50 μm): +0.084 (+0.093 exact). The estimate arrives BELOW the truth, the long stretch with $b_{\text{true}} < 8/9$ outweighing the short one above it; the 09-04 first writing integrated f alone and gave 0.008 – FAIL caught by the independent verification. MEASURED (run_prod_ladder.m, four blocks, 10 seeds, paired to the b_{true}-curve arm): b locked at 8/9: end of descent -0.0030 ± 0.0002 (canon), -0.0005 ± 0.0002 (Meng); hold start -0.0005, -0.0009; the y_2 leg in motion removes $\geq 95\%$ of the open-loop error, sign as derived. b locked at $b_{\text{true}}(\bar{w}_{\text{hold}}) = 0.928$ (oracle): end of descent -0.0066, -0.0017 (more negative: $\hat{b} > b_{\text{true}}$ over the whole mid range, as $f < 0$ says). $\hat{b}$ estimated from 8/9 (production): minus locked 8/9 at the hold start $+0.0014 \pm 0.0009$ (canon), -0.00002 ± 0.00013 (Meng), $\hat{b}$ stays within 0.01 of the seed (0.885 ± 0.017, 0.896 ± 0.021 at the end), $\sqrt{P_{55}}$ $0.039 \to 0.023$, 0.027. Absolute hold level of production with the four blocks: $+0.0012 \pm 0.0016$ (canon), $+0.0004 \pm 0.0023$ (Meng); production as it was (Euler, no mean terms): +0.019, +0.010. OPEN: a paired hold-slope difference constant-b minus curve of -0.176 ± 0.025 (canon), -0.184 ± 0.015 (Meng) $\times 10^{-6}$ per step; not the b'_{true} cross-moment ($< 0.002 \times 10^{-6}$ on the filter's P, probe_lockb_bprime_line.m); with b locked at the hold value it vanishes on canon (-0.006 ± 0.032) but not on Meng (-0.165 ± 0.014), so it is not simply the slope mismatch at the hold height either; -0.0014 over a 5 s hold, mechanism open)

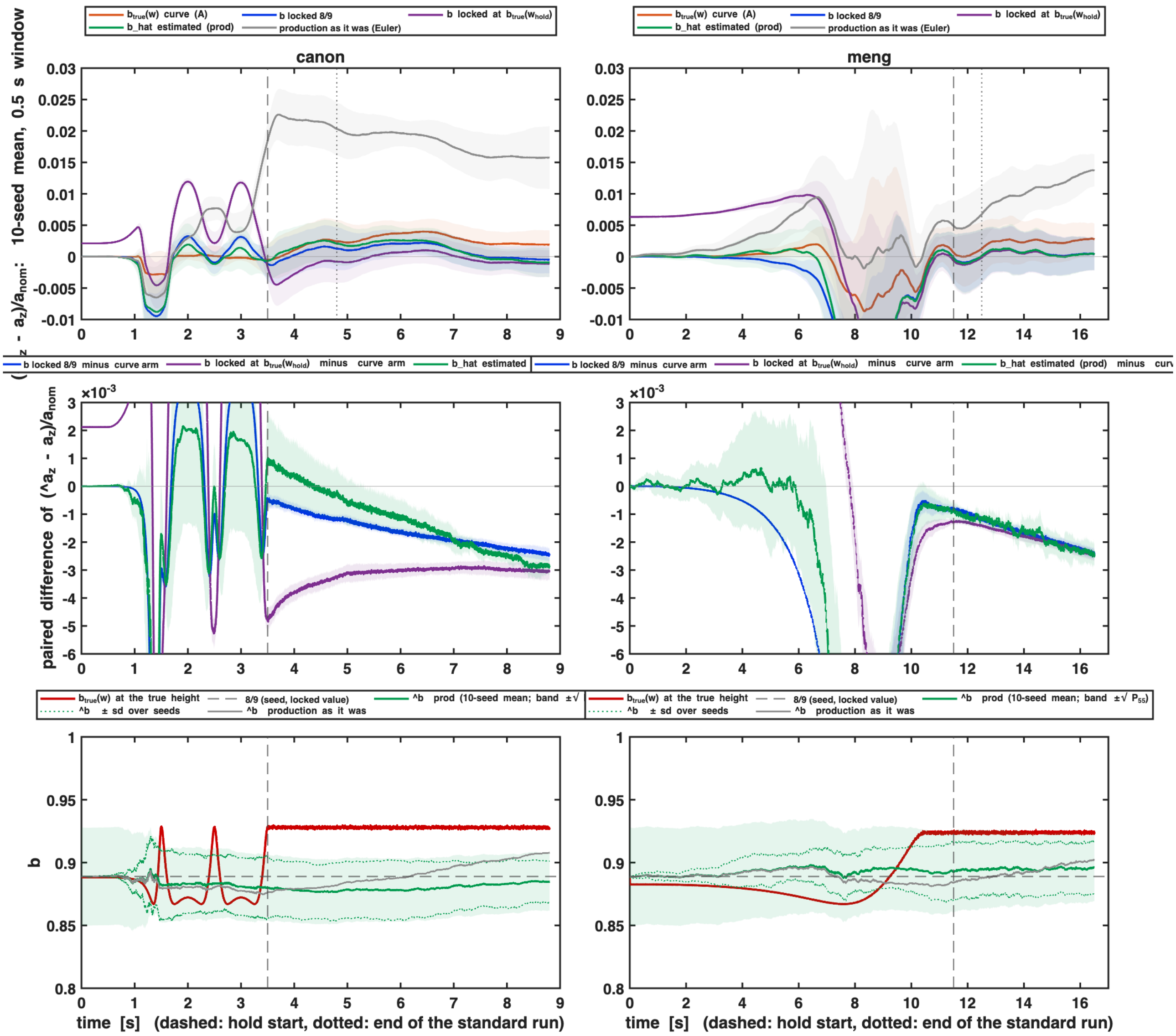
Jacobians and measurement (the entries that change; code: state writing, $\partial\bar{a}'/\partial\hat{b} = +\bar{a}'/\hat{b}$)

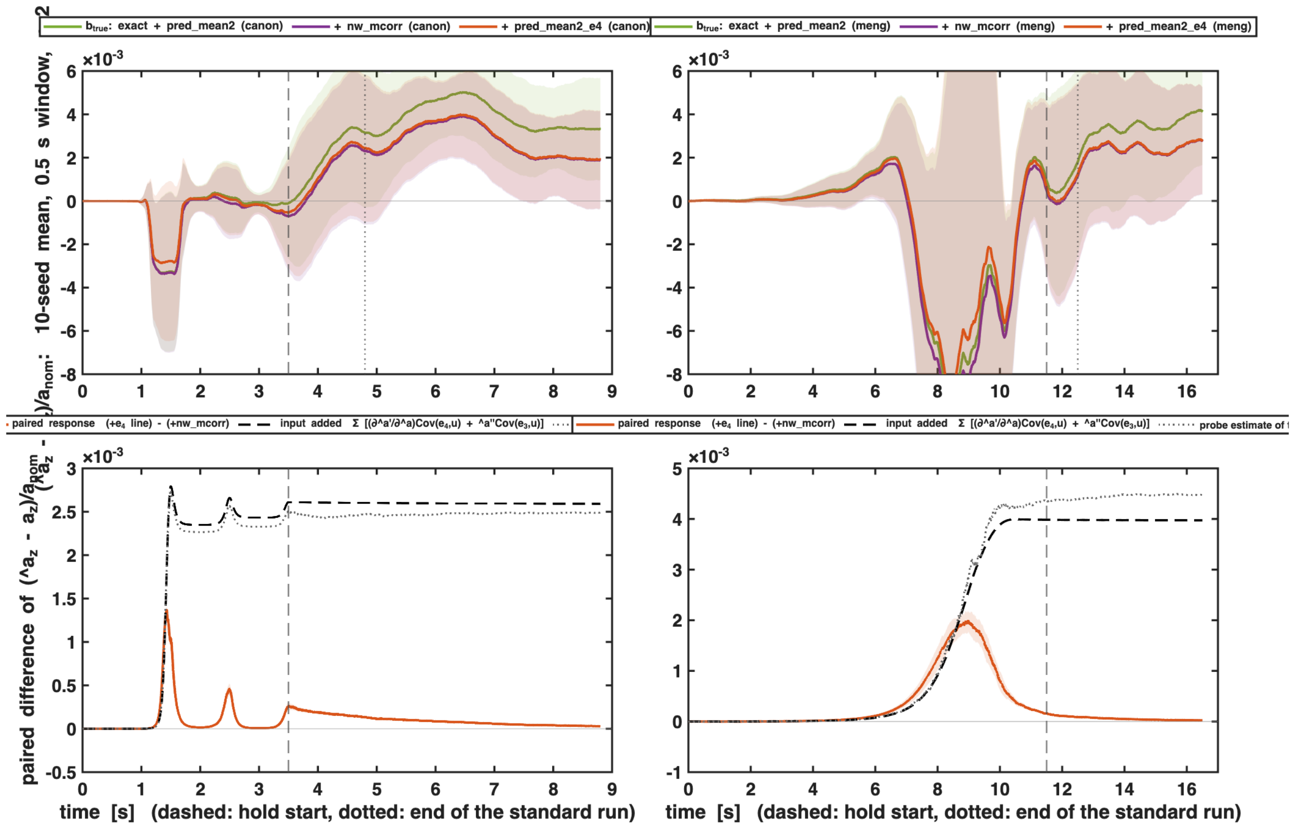
$$\frac{\partial \hat{\bar{a}}_w[k+1]}{\partial \hat{b}[k]} = \frac{\partial \bar{a}'}{\partial \hat{b}}\,\Delta\hat{\bar{w}}[k], \qquad \frac{\partial \hat{b}[k+1]}{\partial \hat{b}[k]} = 1$$

$$y_2[k] = \bar{a}_w[k] - \bar{a}'\,\nabla_d\bar{w}_d[k] + n_a[k], \qquad H_{24} = 1 - \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\nabla_d\bar{w}_d[k], \qquad H_{25} = -\frac{\partial \bar{a}'}{\partial \hat{b}}\,\nabla_d\bar{w}_d[k]$$

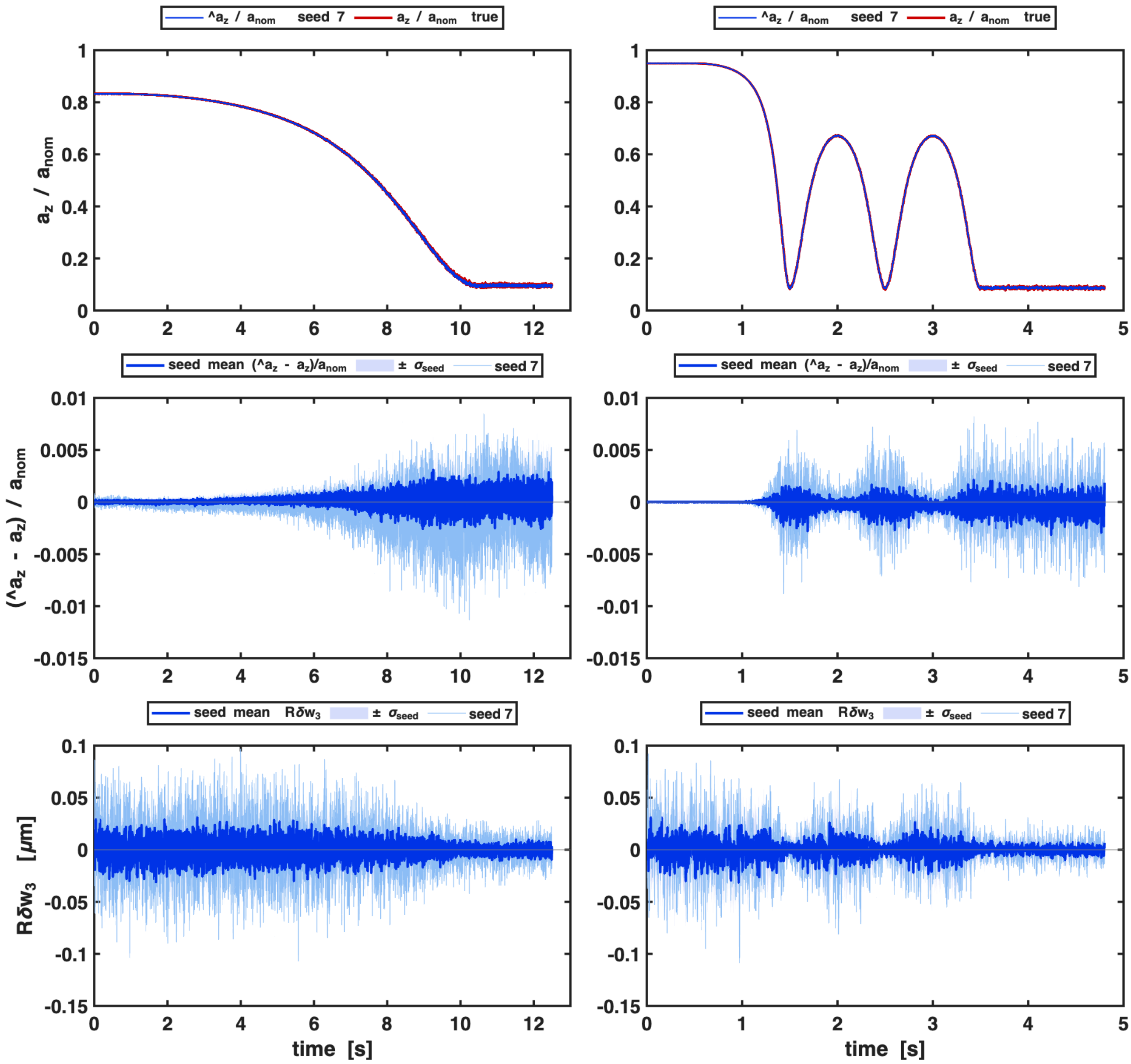
$$g_n[k] = (1-\lambda_c)\,[\,0,\ 0,\ -1,\ \bar{a}',\ 0\,]^\top, \qquad \tilde{Q}_{5j} = \tilde{Q}_{j5} = 0$$

(both $\hat{b}$ columns carry a displacement factor and vanish together in a hold, Section 3 of observability-workflow; locked-b arm: the filter's $P_{i5} = 0$, so the added line vanishes and the model-error lines, including the b'_{true} cross-moment, remain)


canon
meng
$b_{true}(w)$ curve (A)
b_hat estimated (prod)
b locked 8/9
production as it was (Euler)
b locked at $b_{true}(w_{hold})$
10-seed mean, 0.5 s window
b locked 8/9 minus curve arm
b locked at $b_{true}(w_{hold})$ minus curve arm
b_hat estimated (prod) minus curve arm
paired difference of $(\hat{a}_z - a_z)/a_{nom}$
$\times10^{-3}$
$b_{true}(w)$ at the true height
8/9 (seed, locked value)
^b prod (10-seed mean; band $\pm\sqrt{P_{55}}$)
^b ± sd over seeds
^b production as it was
b
time [s] (dashed: hold start, dotted: end of the standard run)



btrue: exact + pred_mean2 (canon)
+ nw_mcorr (canon)
+ pred_mean2_e4 (canon)
btrue: exact + pred_mean2 (meng)
+ nw_mcorr (meng)
+ pred_mean2_e4 (meng)
(^az - az)/anom: 10-seed mean, 0.5 s window, 2
×10-3
paired response (+e4 line) - (+nw_mcorr)
input added Σ [(∂^a'/∂^a)Cov(e4,u) + ^a''Cov(e3,u)]
probe estimate of
paired difference of (^az - az)/anom
time [s] (dashed: hold start, dotted: end of the standard run)



^a_z / a_nom seed 7
a_z / a_nom true
a_z / a_nom
seed mean (^a_z - a_z)/a_nom
± σ_seed
seed 7
(^a_z - a_z) / a_nom
seed mean Rδw_3
Rδw_3 [μm]
time [s]